

An assignement aimed at analyzing an error in a mesurment of a
Biomechanics (Joint force) or Lever system

In biomechanics, a joint lever system describes how bones act as levers, joints act as fulcrums, and muscles provide the effort force to move the body.

$$T = f(F,r,\theta) = F \times r \times \sin(\theta)$$

Where, T is Torque at the joint ,F is Force (muscle or load) ,r is Lever arm distance from the joint , θ is Angle between the force and the lever arm.

EXP	Force	Radius	Angular(degree) Dispalcement	Torque
1	12 ± 0.20	0.30 ± 0.003	85 ± 1.5	3.59 ± 0.07
2	14 ± 0.25	0.28 ± 0.005	87 ± 2	3.9 ± 0.11
3	10 ± 0.15	0.32 ± 0.004	88 ± 1	3.2 ± 0.06
4	16 ± 0.35	0.25 ± 0.006	84 ± 2.5	3.97 ± 0.14
5	13 ± 0.22	0.29 ± 0.003	90 ± 1.2	3.77 ± 0.07
6	15 ± 0.28	0.27 ± 0.004	86 ± 1.8	4.03 ± 0.10
7	11 ± 0.18	0.31 ± 0.005	89 ± 1.3	3.41 ± 0.08
8	17 ± 0.40	0.24 ± 0.006	83 ± 2.2	4.05 ± 0.15
9	12 ± 0.23	0.24 ± 0.006	88 ± 1.7	3.96 ± 0.09
10	12 ± 0.23	0.30 ± 0.005	6 ± 2.1	4.18 ± 0.12

Final Torque = 3.81 ± 0.1

$$T = F \cdot r \cdot \sin \theta \quad \text{where } \delta y = \text{error propagated}$$

$$\delta y = \sqrt{\left(\frac{\partial f}{\partial x_1} \Delta x_1\right)^2 + \left(\frac{\partial f}{\partial x_2} \Delta x_2\right)^2 + \left(\frac{\partial f}{\partial x_3} \Delta x_3\right)^2} = \sqrt{(r \sin \theta \Delta F)^2 + (F \sin \theta \Delta r)^2 + (Fr \cos \theta \Delta \theta)^2}$$

$$\text{Experiment 1, } \delta y_1 = \sqrt{(0.30 \times 0.996 \times 0.20)^2 + (12 \times 0.996 \times 0.003)^2 + (12 \times 0.30 \times 0.087 \times 0.0262)^2}$$

$$= \underline{0.07 \text{ Nm}}$$

$$\text{So } T_1 = \underline{3.59 \pm 0.07 \text{ Nm}}$$

$$\text{Experiment 2, } \delta y_2 = \sqrt{(0.28 \times 0.998 \times 0.25)^2 + (14 \times 0.998 \times 0.005)^2 + (14 \times 0.28 \times 0.052 \times 0.0349)^2}$$

$$= \underline{0.11 \text{ Nm}}, \quad T_2 = \underline{3.90 \pm 0.11 \text{ Nm}}$$

$$\text{Experiment 3, } \delta y_3 = \sqrt{(0.32 \times 0.999 \times 0.15)^2 + (10 \times 0.999 \times 0.004)^2 + (10 \times 0.32 \times 0.035 \times 0.0375)^2}$$

$$\delta y_3 = \underline{0.06 \text{ Nm}}, \quad T_3 = \underline{3.20 \pm 0.06 \text{ Nm}}$$

$$\text{Experiment 4, } \delta y_4 = \sqrt{(0.25 \times 0.995 \times 0.35)^2 + (16 \times 0.995 \times 0.006)^2 + (16 \times 0.25 \times 0.105 \times 0.0436)^2}$$

$$\delta y_4 = \underline{0.14 \text{ Nm}}, \quad T_4 = \underline{3.97 \pm 0.14 \text{ Nm}}$$

$$\text{Experiment 5, } \delta y_5 = \sqrt{(0.29 \times 1 \times 0.22)^2 + (13 \times 1 \times 0.003)^2 + (13 \times 0.29 \times 0.0209)^2}$$

$$\delta y_5 = \underline{0.07 \text{ Nm}}, \quad T_5 = \underline{3.77 \pm 0.07 \text{ Nm}}$$

$$\text{Experiment 6, } \delta y_6 = \sqrt{(0.31 \times 0.997 \times 0.28)^2 + (15 \times 0.997 \times 0.004)^2 + (15 \times 0.27 \times 0.07 \times 0.0314)^2}$$

$$\Rightarrow \delta y_6 = \underline{0.10 \text{ Nm}}, \quad T_6 = \underline{4.03 \pm 0.10 \text{ Nm}}$$

$$\text{Experiment 7, } \delta y_7 = \sqrt{(0.31 \times 1 \times 0.18)^2 + (11 \times 1 \times 0.005)^2 + (11 \times 0.31 \times 0.017 \times 0.0227)^2}$$

$$\delta y_7 = \underline{0.08 \text{ Nm}}, \quad T_7 = \underline{3.41 \pm 0.08 \text{ Nm}}$$

$$\text{Experiment 8, } \delta y_8 = \sqrt{(0.24 \times 0.993 \times 0.4)^2 + (17 \times 0.993 \times 0.006)^2 + (17 \times 0.24 \times 0.122 \times 0.0389)^2}$$

$$\delta y_8 = \underline{0.15 \text{ Nm}}, \quad T_8 = \underline{4.05 \pm 0.15 \text{ Nm}}$$

$$\text{Experiment 9, } \delta y_9 = \sqrt{(0.33 \times 0.999 \times 0.23)^2 + (12 \times 0.999 \times 0.004)^2 + (12 \times 0.33 \times 0.035 \times 0.0297)^2}$$

$$\delta y_9 = \underline{0.09 \text{ Nm}}, \quad T_9 = \underline{3.96 \pm 0.09 \text{ Nm}}$$

Experiment 10 $\delta y_{10} = \sqrt{(0.30 \times 0.998 \times 0.30)^2 + (14 \times 0.998 \times 0.005)^2 + (14 \times 0.30 \times 0.070 \times 0.0366)^2}$

$\Rightarrow \delta y_{10} = 0.12 \text{ Nm} \quad T = 4.18 \pm 0.12 \text{ Nm}$

• Mean Torque $\bar{T} = \frac{T_1 + T_2 + T_3 + \dots + T_{10}}{10}$

$$= \frac{3.59 + 3.90 + 3.20 + 3.97 + 3.77 + 3.03 + 3.41 + 4.05 + 3.96 + 4.18}{10}$$

$= \underline{\underline{3.81 \text{ Nm}}}$

• Standard Deviation $\sigma = \sqrt{\frac{\sum (T_i - \bar{T})^2}{n-1}} = \underline{\underline{0.33 \text{ Nm}}}$ (calculated by the above outputs)

• $SE = \frac{\sigma}{\sqrt{n}} = \frac{0.33}{\sqrt{10}} = 0.10 \text{ Nm}$ $\therefore$ Finally $T = 3.81 \pm 0.10 \text{ Nm}$

• precision (%) = $\frac{\sigma}{\bar{T}} \times 100 \Rightarrow \frac{0.33}{3.81} \times 100 = 8.6\% \rightarrow$ error percentage

$\Rightarrow 100 - 8.6\% = \underline{\underline{91.3\%}}$

• accuracy (%) = $\left(1 - \frac{\bar{T} - T_{\text{theoretical}}}{T_{\text{theory}}}\right) \times 100$

$$= \left(1 - \frac{3.81 - 3.85}{3.85}\right) \times 100$$

$$= \left(1 - \frac{0.04}{3.85}\right) \times 100 = \underline{\underline{98.96\%}}$$

• The percentage error contribution

$\frac{\delta y}{\bar{y}} \Rightarrow$ Force error relative = $\frac{\delta F}{\bar{F}} = \frac{0.256}{13.4} = 0.0191 = 1.91\%$

$\Rightarrow$ Distance = $\frac{\delta r}{\bar{r}} = \frac{0.0044}{0.289} = 0.0152 = 1.52\%$

$\Rightarrow$ Angle = $\theta = 86.6^\circ = \sin(86.6^\circ) = 0.998$, as $\Delta \sin \theta = \cos \theta \Delta \theta$

$= \Delta \sin \theta = 0.059 \times 0.0302$

$= 0.00178$

$\Rightarrow \frac{0.00178}{0.998} = \underline{\underline{0.178\%}}$

$\therefore$ Force contribute the highest error